

Integrated departure and boundary control for low-altitude air city transport systems

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ABSTRACT

Connectivity and digitalization will enable new control measures in urban air mobility operations and open new ways for integrating these measures in real-time traffic management. Hence, new control strategies can be designed to regulate both demand and supply of Low-Altitude Air city Transport (LAAT) systems. This can be achieved by adjusting aircraft departure times, and manipulating transfer aircraft flows at boundary air regions. In this research, new model-based control strategies are designed, where aircraft departure management and boundary control strategies are integrated. The aviation operation can benefit from the proposed flow-oriented control paradigm, which can balance the LAAT system's *supply* and *demand*, i.e. controlling the transfer flow between airspace regions and simultaneously managing the aircraft departure (inflow). The current paper presents the development of different control strategies: Departure Controller (DC), Boundary Controller (BC), and integrated Departure and Boundary Controller (DBC), with supporting simulation results. The designed controllers are tested in a new LAAT framework that considers modeling and control of LAAT operation while capturing the microscopic and macroscopic levels simultaneously.

1. Introduction

In recent years, the development of a new mode of urban air transport, the low-altitude passenger and delivery aircraft, emerges from the advancement of aviation and communication technologies. This advances the concept of urban air mobility (UAM), leading to the invention of new urban air transport systems, namely low-altitude air city transport (LAAT). The literature highlights the growing interest in deploying low-altitude aircraft in the airspace (Doole et al., 2020; Bulusu et al., 2021). Countless aircraft designs are currently in development, and most aircraft are electrically powered, utilize advanced batteries, and are capable of vertical take-off and landing (VTOL) behavior. The VTOL aircraft have the capabilities to fly (vertically or horizontally) and hover depending on the flight mission. These relatively small aircraft can be remotely or autonomously controlled (operated with or without pilots) and may be multi-rotors, fixed-wing, or tilt-wing. This paper collectively refers to these aircraft as low-altitude aircraft. LAAT systems will include the low-altitude aircraft transferring passengers and goods in urban areas using low-altitude levels of urban airspace.

LAAT operations can utilize urban transport networks by including a third-dimensional space for additional mobility alternatives. However, several new operational issues and questions are being raised, see Bulusu (2019), Straubinger et al. (2020), Bauranov and Rakas (2021) and Cohen et al. (2021). One concern is that traffic congestion may be formed in a constrained urban airspace, as the

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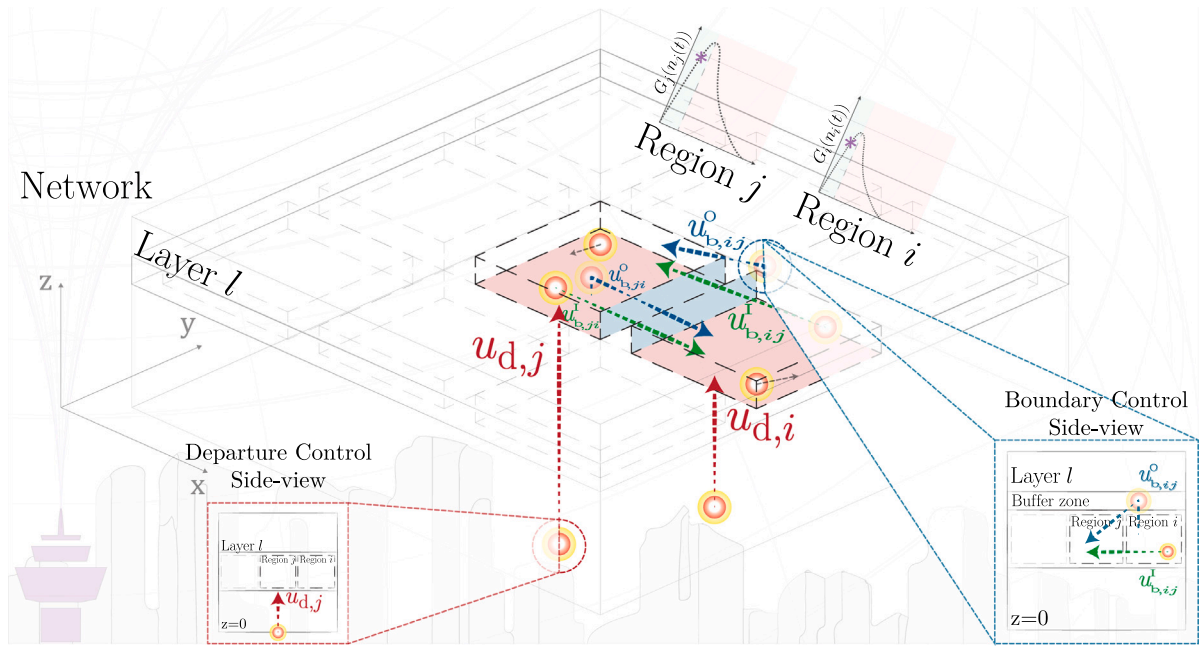


Fig. 1. A schematic digitalized LAAT network consists of layers ($l \in \mathcal{L}$) divided into regions ($i, j \in \mathcal{R}$), with integrating departure control inputs $u_{d,j}, u_{d,i}$, and boundary control inputs $u_{b,i,j}^l, u_{b,j,i}^l, u_{b,j,i}^0, u_{b,i,j}^0$.

number of LAAT aircraft grows. Moreover, in current aviation operations, aircraft are individually controlled under rigid individual operations with limited capacity and safety constraints, while future aircraft can be aggregately controlled under flexible aircraft flow operations. Given the high complexity of individual movements of aircraft in the 3D space, traffic control strategies can be designed with a hierarchical control approach, to control flows of aircraft in specific regions within the network by utilizing the concept of Macroscopic Fundamental Diagram (MFD). While there are some similarities between urban ground traffic systems and LAAT systems, there are also additional challenges that require revisiting both the network modeling and control based on MFD dynamics.

The operation of LAAT traffic in the era of automation and communication differs greatly from that of ground traffic. The key flow-oriented differences between ground traffic and air traffic at the aggregation level of individual entity (vehicle or aircraft) are divided into two main challenges: modeling and control. In modeling, the direction and dimensionality are extended, i.e. the aircraft have multi-directional flows traveling in the 3rd dimension. In control, the aircraft interactions are more complex, especially in collision avoidance, while the safety constraints are very restrictive. In addition, all LAAT aircraft can be fully controlled, as an unmanned or manned aircraft must follow the given commands (to change route or velocity) by the control center. Traffic urban management of LAAT traffic can coordinate and integrate their controlled subsystems, as they are expected to be based on digital technology and telecommunications, see [Straubinger et al. \(2020\)](#).

Advanced technologies are enabling the futuristic aviation ecosystem. Currently, expert humans are responsible for most safety critical tasks, which represent the lowest maturity levels. In the future, aircraft operations and airspace management are expected to be essentially autonomous, which represent the highest maturity levels with high density airspace. The UAM Maturity Level (UML) was recently presented in [Goodrich and Theodore \(2021\)](#) which is aligned with this work perspective that advanced traffic control strategies can be relied on the upcoming technology.

Connectivity and digitalization will enable new control measures in aviation operations and open new ways for integrating these measures in real time urban traffic management. Hence, new control strategies can be designed to regulate LAAT demand and supply. This can be achieved by manipulating aircraft departures, aircraft routings, aircraft speed, etc. In this paper, new model-based control strategies are designed, where aircraft departure management and boundary control are *integrated*, as depicted in [Fig. 1](#). While the urban ground systems are still limited in controllability, the aviation operation can benefit from the proposed flow-oriented control paradigm, which can balance the LAAT system's *supply* and *demand*. It should be noted that LAAT controlled systems might create system queues, which have a high impact on the system's efficiency and can lead to environmental and economical challenges. Therefore, it is worth investigating the interaction and synchronization of controlling both the *input* (demand) and *transfer* traffic flows (supply) of LAAT systems. The investigation focuses on balancing the constraints and capabilities of a LAAT system to achieve system stability. To the authors' knowledge, integrated *departure management* and *boundary control* strategies for LAAT systems have never been investigated.

Note that in urban ground systems, several perimeter control strategies have been introduced in the literature [Daganzo \(2007\)](#), [Geroliminis and Daganzo \(2008\)](#), [Keyvan-Ekbatani et al. \(2012\)](#), [Haddad and Geroliminis \(2012\)](#), [Haddad \(2017b\)](#), [Kouvelas et al.](#)

(2017), Yildirimoglu and Ramezani (2019), Ingoles et al. (2020), Sirmatel et al. (2021), Batista et al. (2022) and Sirmatel and Yildirimoglu (2023). Although the proposed boundary control in this paper for LAAT systems has the same concept of the perimeter control in urban ground traffic, i.e. metering the flows transferring from one region to another, yet they are implemented differently due to the different dynamics of vehicles and aircraft (such as queuing vs. hovering, traveling in two-spatial-dimensional flow vs. three-spatial-dimensional flow, etc.) and the different structural configurations of the networks (2D vs. 3D). In the case of perimeter control in ground systems, vehicles can be metered by traffic signals and wait to transfer to the next region by queuing on roads at the perimeter. On the other hand, in the case of the boundary control in LAAT systems, in this paper we design the boundary control while leveraging the 3rd spatial dimension. Controlled aircraft are directed to move to a buffer zone above the region (within the same layer) and wait there by hovering, as illustrated in the enlarged right box in Fig. 1. While waiting in the buffer zone, located far from the boundary, can improve the operational performance of the boundary, it may introduce other issues that could potentially degrade the traffic performance, such as an increase in the number of conflicts between aircraft and higher energy (battery) consumption. It should be noted that aggregate boundary queue dynamics have been investigated in Haddad (2017b) or Ni and Cassidy (2020) for urban ground systems, yet introducing them for LAAT systems imposes new dynamics and control issues that should be addressed both at the microscopic behavior (e.g., collision avoidance) and at the macroscopic level in terms of controlling the flows entering and exiting the buffer (or exiting a region and entering a region).

Traffic flow models are needed to develop integrated strategies for LAAT systems. Following the operational concept of controlling flows of aircraft traffic, a modeling approach from the traffic flow theory can be adopted. The literature shows that the Macroscopic Fundamental Diagram (MFD) is a powerful modeling approach for understanding and managing vehicular networks from a theoretical as well as an application-oriented perspective. The MFD aggregates traffic information from the whole network and states traffic variables that inform us of the network condition (free flow or congestion). MFD as a concept was first proposed in Godfrey (1969), revisited by Mahmassani et al. (1987) and Daganzo (2007), while it was experimentally proven to exist for urban areas in Geroliminis and Daganzo (2008). The interested readers can refer to Haddad (2017b), Ni and Cassidy (2020), Sirmatel et al. (2021), Johari et al. (2021) and Menelaou et al. (2022) for previous MFD works. In terms of the dynamics representation of the MFD-based model for a single region, one can refer to Small and Chu (2003) and Daganzo (2007) and for multi-region systems to Ramezani et al. (2015) and many other studies. The advantages of utilizing an MFD-based model to solve the LAAT control problem is that the complexity is reduced by aggregating the information and control actions at the macroscopic level (network-level), and simultaneously transforming the control actions to the microscopic level (vehicle-level). Such hierarchical modeling and control approach is preferable compared to a global control approach, where only the microscopic level is being controlled directly. Compared to a global control approach, hierarchical modeling and control approaches in large-scale traffic networks allow for a reduction in the number of states and control inputs in the control problem, and increase the space for a feasible set of optimization problems. We believe that it is theoretically possible to develop collective and aggregate aircraft traffic flow models for LAAT systems, and based on that, to design feedback traffic control strategies.

There has been very limited exploration of using traffic flow models for LAAT systems in the literature. The studies in Battista and Ni (2017), Ivan Tereshchenko and Zou (2020), Cummings and Mahmassani (2021, 2022), Aarts et al. (2023), Hoogendoorn et al. (2023) and Cummings and Mahmassani (2024) focused on the macroscopic level for LAAT systems, and showed a similar behavior of MFD shape to ground traffic, however, only the free flow conditions are presented. In Haddad et al. (2021), a congested regime is shown in the MFD curve while considering the virtual queues that surround the airspace. While in Safadi et al. (2023a), different case study examples were simulated to analyze the full shape of MFD curves for LAAT systems, considering different algorithms, aircraft, and airspace settings, where a congested regime is sufficiently presented.

Recently, an adaptive boundary control strategy considering the aggregated behavior of aircraft flows in large-scale LAAT systems was developed in Haddad et al. (2021). The strategy was designed to operate a LAAT network at a critical or desired accumulation state, generated from a reference model. The control strategy manages the transfer of aircraft, i.e. the output traffic LAAT flow, at the boundaries of the urban regions. It should be stressed that in Haddad et al. (2021) only the boundary control was considered; and the control-oriented model and the plant model were the same developed aggregated MFD-based model, i.e. the lower level complexity of collision avoidance and individual aircraft interaction were not considered. Finally, although an aircraft departure management for LAAT system was designed in Safadi et al. (2023b), only the departure control was examined with a simplified MFD-based model, which does not include the transfer flows between regions within the network.

This paper investigates multi-regional model-based traffic management for LAAT systems. Feedback control strategies are developed based on a new accumulation-based model as a means to manage the *input* and *transfer* traffic flows in the airspace. Hence,

- (i) a new aggregate control-oriented LAAT model is formulated in this paper, serving as a control model for designing the integrated departure and boundary control strategy. The new dynamics model the aggregated behavior of aircraft flows in LAAT operation, while utilizing advanced control strategies, and capture the system queuing behavior;
- (ii) an original feedback control strategy is designed in this paper, which integrates departure management and boundary control methods. The real-time strategy is designed to minimize the system delays, including departure and boundary delays. The control and plant models are different, i.e. the control model is based on macroscopic aggregated behavior, while the plant model is based on microscopic individual aircraft behavior. The developed feedback control strategies at the macroscopic level are transformed to be implemented at the microscopic level using an aircraft command subsystem and its algorithms, in an attempt to verify the optimality of the airspace conditions. The original development of the aircraft command subsystem and its algorithms required major enhancements in the application of the microscopic plant model to accommodate and verify advanced traffic control strategies and the proposed hierarchical modeling and control solutions;

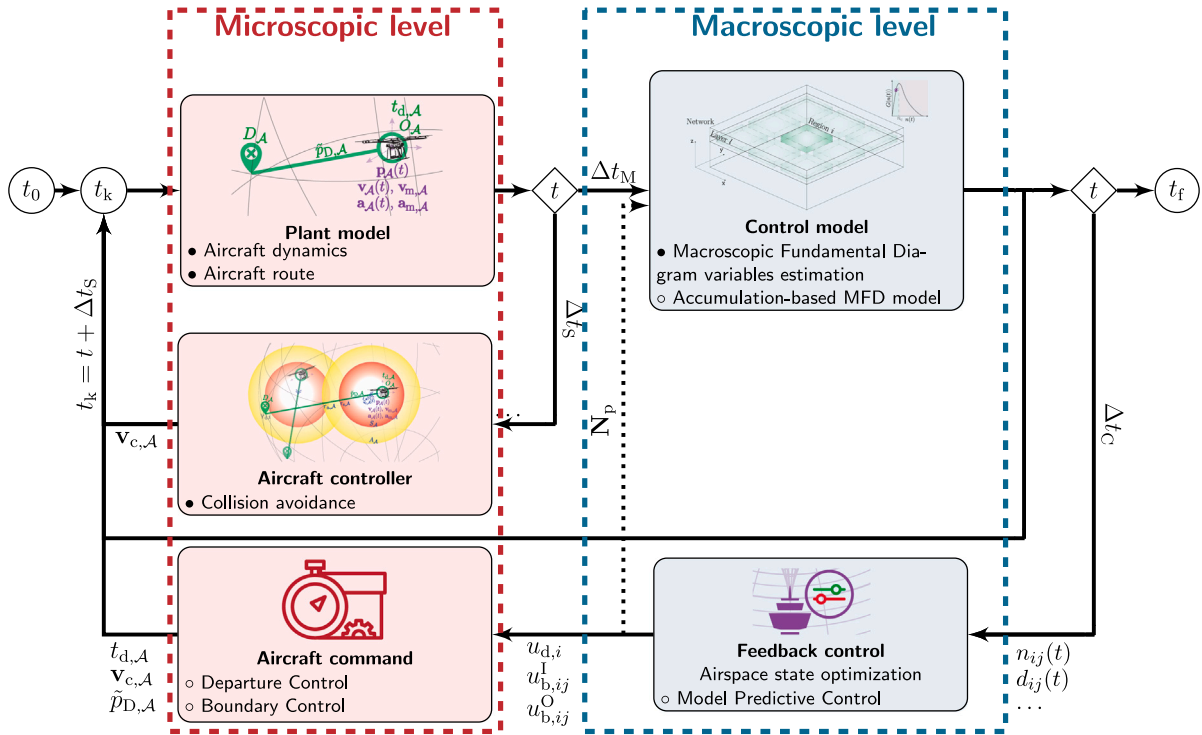


Fig. 2. LAAT model-based operational framework with departure and boundary control, while combining the microscopic and macroscopic levels of LAAT controlled systems.

(iii) a novel model-based traffic management framework is constructed, which couples modeling and control at the microscopic and macroscopic levels, capturing LAAT aircraft and airspace dynamics and characteristics. The presented LAAT framework that includes the implementation of traffic control strategies is used to examine the strategies in different cases, by testing the departure management and boundary control separately, and as an integrated strategy.

2. Developing LAAT aircraft departures and boundary flows model with MFD-based dynamics for integrated control design

2.1. LAAT model-based operational framework

A new LAAT framework that couples modeling and control of LAAT systems is developed, as shown in Fig. 2. The developed framework integrates the two aggregation levels, i.e. microscopic and macroscopic.

In the presented framework, the microscopic model is utilized as a plant model, which describes the “reality” with detailed aircraft behavior, while the macroscopic model is utilized as a control model to design the control inputs. The traffic LAAT dynamics of the microscopic level steer the aggregate dynamics of the macroscopic level, which its characteristics variables are inserted as an input to the control model. Simultaneously, a feedback control strategy is designed to operationally optimize the airspace state following the airspace dynamics and identification. Then, the calculated optimal control inputs are transferred and transformed to the microscopic level via an aircraft command subsystem that optimizes the aircraft departure times and controls the aircraft at the regional boundary.

The presented framework allows us to investigate different control strategies while utilizing airspace dynamics. The goal is to study three strategies: (i) departure management, i.e. departure time optimization, (ii) boundary control, i.e. controlling the aircraft prior to flying through a virtual boundary; and (iii) integrated departure and boundary control.

2.2. Microscopic model for air traffic flow

To model LAAT operation sufficiently, a realistic plant model that captures individual aircraft movements inside the airspace is needed. We adopt the developed microscopic model in Safadi et al. (2023a), which can capture the dynamics of the aircraft while maintaining collision-free fly. A brief description of the microscopic model is given here as follows, while for a detailed description, the interested reader is referred to the original paper. Based on this model, we construct the MFD for LAAT networks to capture the network dynamics. Moreover, the microscopic model serves as a plant model, i.e. reality, to test the traffic control strategies.

There are three main parts in the microscopic LAAT model: (1) aircraft dynamics, (2) aircraft route, and (3) collision avoidance. The aircraft dynamics are formed according to the kinematic behavior where the states are position $\mathbf{p}_A(t)$ and velocity $\mathbf{v}_A(t)$, and the control input is acceleration $\mathbf{a}_A(t)$ for an aircraft A with a motion in 3D (X-Y-Z) space. Maximum velocity $\mathbf{v}_{m,A}$ and acceleration $\mathbf{a}_{m,A}$ bounds are set for each aircraft. The aircraft route determines individual aircraft routes, where each aircraft departs at a predefined time $t_{d,A}$ from a predefined origin point O_A to a predefined destination point D_A . A direct line between the two points is set as the original route $\tilde{p}_{D,A}$. It is important to note that the final route depends on the collision avoidance algorithm, which should be set. A collision avoidance algorithm should control the aircraft to avoid collisions with other aircraft or obstacles. To control LAAT aircraft, two types of space around each aircraft A are defined: (i) safety space S_A , and (ii) avoidance space A_A . The safety space S_A is determined to avoid a conflict and the avoidance space A_A is determined to start avoidance control. The aircraft avoids a collision with a predefined safety radius $r_{s,A}$ and avoidance radius $r_{a,A}$, by determining the aircraft control input $\mathbf{v}_{c,A}$ using an Artificial Potential Field (APF) approach based on Lyapunov-Like functions. In addition, following a decentralized control approach, a detection space for each aircraft is defined. The detection space $\mathcal{D}_A$ of aircraft A is centered in its position $\mathbf{p}_A(t)$ with the detection radius $r_{d,A}$. The aircraft's objective is to follow predefined waypoints and avoid collision with moving obstacles.

The aircraft model has two properties: (i) conflict-free, i.e. even if an aircraft enters into the safety space of another aircraft, it can keep away from the neighboring aircraft rapidly; and (ii) convergence to desired waypoints is guaranteed. Moreover, regarding practicability, the developed control has three features: (i) a practical motion model is used, i.e. a double integrator model with a velocity command as an input is proposed for aircraft; (ii) detection of neighboring aircraft IDs of an aircraft is not required by the control; and (iii) control saturation is imposed, i.e. the maximum velocity command in the controller is restricted according to a priori determined requirement.

2.3. Macroscopic fundamental diagram for LAAT networks

By aggregating the aircraft trajectories in the network, it is possible to identify the macroscopic level variables to capture the airspace traffic conditions and to construct the MFD. We adopt the spatial structure that divides the airspace into several vertical latitudes and each latitude into different regions. The MFD concept can be determined for the whole network or per region, where the boundaries of each region need to be determined to evaluate the macroscopic traffic flow variables. Furthermore, as shown in Fig. 2, the MFD variables are calculated during a specific time interval Δt_M , i.e. each time interval Δt_M the aircraft trajectories are analyzed to determine the macroscopic traffic flow variables.

The main traffic characteristics and physics laws of vehicular traffic flow theory can be generalized to aircraft traffic flow by adapting the one-spatial-dimensional flow to three-spatial-dimensional flow. We follow the generalized definitions of Edie (1963) to calculate the MFD variables. Using the aircraft trajectories, the travel time and distance of each aircraft for a fixed time period are measured and aggregated, to calculate the main macroscopic traffic flow variables, i.e. flow $Q(t)$, density $K(t)$, and speed $V(t)$. In addition, other macroscopic traffic flow variables are calculated such as accumulation $n(t)$, outflow $G(t)$, and production $P(t)$, which are evaluated according to the regions and network boundaries and the position of aircraft. These macroscopic traffic flow variables enable the development of an accumulation-based model. Moreover, to determine the airspace performance, additional factors are estimated such as: queues and delays. It is common in the literature on traffic systems, e.g. Papageorgiou et al. (2003) and Haddad (2017a), to compare the total time spent in the system when investigating different strategies which can be also used here to reflect accurately the LAAT system performance by aggregating travel time and waiting time of aircraft.

By estimating the above macroscopic traffic flow variables, one can identify macroscopically the airspace state, i.e. if the airspace is in free flow or congestion conditions. This estimation allows to model the airspace dynamics and then to develop a control model for the design of a feedback control strategy. The development of the control model is described in Section 2.4.

2.4. Departures and boundary flows model

Following the introduced LAAT model-based operational framework, a macroscopic model allows us to capture the airspace conditions and to model the aggregated behavior of aircraft flows in LAAT network. The proposed model, namely *departures and boundary flows model*, can manipulate the departure of aircraft (inflow rate) and control the transfer flows between neighboring regions. The model can be utilized as a control model to apply traffic control strategies, in order to optimize the airspace conditions and operate LAAT system efficiently.

For this reason, a new MFD-based model is developed, where (i) the demand that enters region i is being controlled by a regional (aggregate) control input, namely *departure controller (DC)*, $u_{d,i} [-]$ ($i \in \mathcal{R}$, where $\mathcal{R}$ is a set containing all regions in the network). The departure control input $u_{d,i}$ determines the ratio of the inflow that enters region i , i.e. manipulating the demand; and (ii) the transfer flow from region i to region j is being controlled by a *boundary controller (BC)*, with boundary control inputs $u_{b,ij}^I, u_{b,ij}^O [-]$ ($i, j \in \mathcal{R}_{ij}$, where $\mathcal{R}_{ij}$ is a set containing all neighboring regions at each region i with destination to another region j). The boundary control input $u_{b,ij}^I$ determines the ratio of the transfer flow from region i to neighboring region j that is allowed to pass the boundary ij , i.e. flying *inside* the neighboring region, where the complementary proportion of the transfer flow $(1 - u_{b,ij}^I)$ is assigned to wait (hover) at the boundary ij outside the region (a dedicated buffer zone), i.e. assigning traveling aircraft, prior to passing a boundary, to wait for approval to pass, therefore creating aircraft boundary queues. Additionally, the boundary control input $u_{b,ij}^O [-]$ determines the ratio of the discharging aircraft *out* of the boundary queue of region i to neighboring region j , and the complementary proportion $(1 - u_{b,ij}^O)$ is assigned on aircraft to wait for approval to pass the boundary. Note that the actuation and implementation of the control inputs are explained later in detail in Section 2.5.

In this section, we elaborate on the proposed accumulation-based (MFD-based) model, where (i) the transfer flow interactions between regions i, j are captured, i.e. a multi-region model for centralized control approach, (ii) the departure queue dynamics are modeled, i.e. modeling the storage of aircraft in the take-off facilities (airports or vertiports), and (iii) the boundary queue dynamics are modeled, i.e. modeling the storage of queuing aircraft in the boundary within the dedicated buffer zones. Note that the model is given for a one-layer multi-region network, and it can be applied to multi-layer multi-region networks sufficiently with the same modeling methodology.¹

Remark 1. The definitions of departure and boundary queues in the paper are referred to the accumulated number of queued aircraft, on the ground or at the boundary, respectively. Accordingly, the queue dynamics are the derivatives over time of the aircraft queuing.

The accumulation-based model was formulated for a single region based on the mass conservation equation for vehicles, see [Small and Chu \(2003\)](#) and [Daganzo \(2007\)](#), and recently was introduced for aircraft flows in [Haddad et al. \(2021\)](#). Considering one-region network, the dynamics of accumulation of traveling aircraft $n(t)$ [aircraft], i.e. $\dot{n}(t)$ [aircraft/s], is formulated as follows,

$$\dot{n}(t) = d(t) - G(n(t)), \quad (1)$$

where $d(t)$ [aircraft/s] is the inflow rate (demand) and $G(n(t))$ [aircraft/s] is the outflow rate (supply) in the system.

Simulation results in [Safadi et al. \(2023a\)](#) show that the LAAT MFD shape $G(n)$ [aircraft/s] can be approximated by a non-symmetric unimodal curve for a subset of LAAT network, where the critical accumulation n_c [aircraft] is the number of aircraft that allows flying freely, achieving the maximum outflow point G_m [aircraft/s] without reaching congestion. Then, the outflow rate $G(n(t))$ [aircraft/s] is an estimated function $G(n(t)) = (1/a_1) \cdot n(t) \cdot \exp(-1/a_2) \cdot (n(t)/a_3)^{a_2}$, where a_1, a_2, a_3 are the estimated parameters from curve fitting.

In this study, we construct the model to manipulate both the demand and supply in the modeling. First, let us start with the demand. In order to impose a departure control, one can include a control input $u_d(t)$ [-] which determines the ratio of the inflow rate $d(t)$ [aircraft/s], then a general dynamics can be formulated as follows,

$$\dot{n}(t) = d(t) \cdot u_d(t) - G(n(t)). \quad (2)$$

When considering a multi-region network, Eq. (2) is modified to include several departure control inputs, where $u_{d,i}$ [-] determines the ratio of the inflow in each region i .

Second, we are interested to impose a boundary control between regions, then the accumulation of aircraft traveling between neighboring regions needs to be modeled in the dynamics. Accordingly, one can include a control input $u_{b,ij}(t)$ [-] which determines the ratio of the outflow rate in region i with destination to neighboring region j , i.e. $b_{ij}(t) \cdot G_i(n_i(t))$ with $b_{ij}(t)$ [-] as a weighting factor which is assumed to have a general nonlinear form, then one gets

$$\dot{n}_{ij}(t) = d_{ij}(t) - b_{ij}(t) \cdot G_i(n_i(t)) \cdot u_{b,ij}(t), \quad (3)$$

where $n_{ij}(t)$ [aircraft] is the accumulation of aircraft at time t in region i with destination to neighboring region j , and correspondingly the time derivative is denoted by $\dot{n}_{ij}(t)$ [aircraft/s], $d_{ij}(t)$ [aircraft/s] is the inflow rate (demand) at time t in region i with destination to region j .

Finally, one can integrate both the departure and boundary controllers. Let us distinguish between *traveling* and *queuing* aircraft. Then, the accumulation of *traveling* aircraft at time t in region i with destination to neighboring region j is denoted by $n_{ij,T}(t)$ [aircraft], and the time derivative of the accumulation $\dot{n}_{ij,T}(t)$ [aircraft/s] with departure and boundary control is modeled as follows,

$$\dot{n}_{ij,T}(t) = d_{ij}(t) \cdot u_{d,i}(t) - b_{ij}(t) \cdot G_i(n_i(t)) \cdot u_{b,ij}^I(t), \quad \forall i, j \in \mathcal{R}_{ij}. \quad (4)$$

The departure and boundary control affects also the dynamics of accumulation of *traveling* aircraft within the regions. Let us denote the accumulation of *traveling* aircraft at time t in region i with destination i by $n_{ii,T}(t)$ [aircraft], then the time derivative of the accumulation $\dot{n}_{ii,T}(t)$ [aircraft/s] is modeled as follows,

$$\dot{n}_{ii,T}(t) = d_{ii}(t) \cdot u_{d,i}(t) + \sum_{j \in \mathcal{R}_{ij}} b_{ji}(t) \cdot G_j(n_j(t)) \cdot u_{b,ji}^I(t) + \sum_{j \in \mathcal{R}_{ij}} b_{ji}^Q(t) \cdot S_j^Q \cdot u_{b,ji}^O(t) - b_{ii}(t) \cdot G_i(n_i(t)), \quad \forall i \in \mathcal{R}, \quad (5)$$

where $b_{ii}(t), b_{ji}(t), b_{ji}^Q(t)$ [-] are weighting factors which are assumed to have general nonlinear forms, and S_j^Q [aircraft/s] is the saturation flow at the boundary in region j , i.e. the maximum discharge aircraft rate from the boundary queue. In other words, (5) includes four terms: the departing aircraft within the region, the aircraft allowed to pass the boundary, the aircraft discharged from the boundary queues, and the outflow in the region as aircraft finishing their trips within the region.

Therefore, the total accumulation of traveling aircraft in each region i is denoted by $n_i(t)$, and the time derivatives $\dot{n}_i(t)$ are modeled as follows,

$$\dot{n}_i(t) = \dot{n}_{ii,T}(t) + \sum_{j \in \mathcal{R}_{ij}} \dot{n}_{ij,T}(t), \quad \forall i \in \mathcal{R}. \quad (6)$$

¹ Note that special attention needs to be given to the traffic assignment and routes within the network to determine the distribution of demand and flows among the layers and regions within the airspace. Please refer to [Haddad et al. \(2021\)](#) for further details on MFD-based model for a multi-layer case.

Moreover, the model includes the *departure* queue dynamics, which describe the number of aircraft waiting on the ground (outside the network) to enter the airspace regions. The departure queue at time t in region i is denoted by $n_{i,DQ}(t)$, and according to the demand in the region and the departure control inputs, the time derivative of the departure queue $\dot{n}_{i,DQ}(t)$ is modeled as follows,²

$$\dot{n}_{i,DQ}(t) = (d_{ii}(t) + \sum_{j \in \mathcal{R}} d_{ij}(t)) \cdot (1 - u_{d,i}(t)), \quad \forall i \in \mathcal{R}. \quad (7)$$

Additionally, the model includes the *boundary* queue dynamics, which describe the number of aircraft waiting to cross the regions on the boundary outside the network (or region), i.e. aircraft are queuing outside the network (or region). The boundary queue at time t in region i with destination to neighboring region j is denoted by $n_{ij,Q}(t)$. According to the outflow, the discharge of queues on the boundary and the boundary control inputs, the time derivative of the departure queue $\dot{n}_{ij,Q}(t)$ is modeled as follows,

$$\dot{n}_{ij,Q}(t) = (b_{ij}(t) \cdot G_i(n_i(t))) \cdot (1 - u_{b,ij}^I(t)) - b_{ij}^O(t) \cdot S_i^O \cdot u_{b,ij}^O(t), \quad \forall i, j \in \mathcal{R}_{ij}. \quad (8)$$

Recall that $u_{b,ij}^I(t)$ determines the transfer flow from region i to neighboring region j which applies the queuing behavior at the boundary ij , and $u_{b,ij}^O(t)$ determines the released boundary queue flow from region i to neighboring region j .

The LAAT system aims to operate efficiently while maintaining safety and resilience. In this paper, the objective function $J(\star)$ [aircraft · s] is chosen to minimize delays of LAAT operations under different demand conditions. The introduced objective function consists of three terms, which can be written as follows

$$J(\star) = J_1 + J_2 + J_3, \quad (9)$$

where J_1 considers the traveling aircraft delays in the system with weighting factors ω_{n_i} [1/aircraft] for each region i ($\forall i \in \mathcal{R}$), which is reflected by

$$J_1(n_{ij}(t)) = \int_0^{t_f} \left(\sum_{i \in \mathcal{R}} \omega_{n_i} \cdot \|n_i(t)\|^2 \right) dt. \quad (10)$$

Moreover, J_2 considers the departure queue delays in the system by minimizing the error between the departure control inputs and corresponding control reference values $u_{r,d,i}$ [-] with weighting factors $\omega_{u_{d,i}}$ [aircraft] for each region i ($\forall i \in \mathcal{R}$), which is reflected by

$$J_2(u_{d,i}(t)) = \int_0^{t_f} \left(\sum_{i \in \mathcal{R}} \omega_{u_{d,i}} \cdot \|u_{r,d,i} - u_{d,i}(t)\|^2 \right) dt, \quad (11)$$

J_3 considers the boundary queue delays at the regional boundaries by minimizing the errors between the control inputs (of the input and output flows of the queues) $u_{r,b,ij}^I$ [-], $u_{r,b,ij}^O$ [-] and their corresponding reference values $u_{r,b,ij}^I$ [-], $u_{r,b,ij}^O$ [-], respectively; with weighting factors $\omega_{u_{b,ij}^I}$, $\omega_{u_{b,ij}^O}$ [aircraft] for each boundary ij ($\forall i, j \in \mathcal{R}_{ij}$). This reads as

$$J_3(u_{b,ij}^I(t), u_{b,ij}^O(t)) = \int_0^{t_f} \left(\sum_{i,j \in \mathcal{R}_{ij}} \omega_{u_{b,ij}^I} \cdot \|u_{r,b,ij}^I - u_{b,ij}^I(t)\|^2 \right) dt + \int_0^{t_f} \left(\sum_{i,j \in \mathcal{R}_{ij}} \omega_{u_{b,ij}^O} \cdot \|u_{r,b,ij}^O - u_{b,ij}^O(t)\|^2 \right) dt. \quad (12)$$

The terms are weighted by the factors ω_{n_i} [1/aircraft], $\omega_{u_{d,i}}$ [aircraft], and $\omega_{u_{b,ij}^I}$, $\omega_{u_{b,ij}^O}$ [aircraft], respectively.

Therefore, the optimal combined departure and boundary control problem is formulated as follows,

$$\begin{aligned} \min_{\substack{u_{d,i}(t) \\ u_{b,ij}^I(t) \\ u_{b,ij}^O(t)}} J(\star) = & \int_0^{t_f} \left(\sum_{i \in \mathcal{R}} \omega_{n_i} \cdot \|n_i(t)\|^2 + \sum_{i \in \mathcal{R}} \omega_{u_{d,i}} \cdot \|u_{r,d,i} - u_{d,i}(t)\|^2 \right. \\ & \left. + \sum_{i,j \in \mathcal{R}_{ij}} \omega_{u_{b,ij}^I} \cdot \|u_{r,b,ij}^I - u_{b,ij}^I(t)\|^2 + \sum_{i,j \in \mathcal{R}_{ij}} \omega_{u_{b,ij}^O} \cdot \|u_{r,b,ij}^O - u_{b,ij}^O(t)\|^2 \right) dt \end{aligned} \quad (13)$$

s.t.

Eqs. (4)–(8),

$$0 \leq n_i(t) \leq \bar{n}_i, \quad \forall i \in \mathcal{R}, \quad (14)$$

$$0 \leq n_{i,DQ}(t) \leq \bar{n}_{i,DQ}, \quad \forall i \in \mathcal{R}, \quad (15)$$

$$0 \leq n_{ij,Q}(t) \leq \bar{n}_{ij,Q}, \quad \forall i, j \in \mathcal{R}_{ij}, \quad (16)$$

$$\underline{u}_{d,i} \leq u_{d,i}(t) \leq \bar{u}_{d,i}, \quad \forall i \in \mathcal{R}, \quad (17)$$

$$\underline{u}_{b,ij}^I \leq u_{b,ij}^I(t) \leq \bar{u}_{b,ij}^I, \quad \forall i, j \in \mathcal{R}_{ij}, \quad (18)$$

$$\underline{u}_{b,ij}^O \leq u_{b,ij}^O(t) \leq \bar{u}_{b,ij}^O, \quad \forall i, j \in \mathcal{R}_{ij}, \quad (19)$$

² The airports or vertiports are assumed to be in saturated or oversaturated conditions during rush hours. Such assumption is imposed to simplify the equation, while other general mass conservation equation for aircraft at airports can be suggested without altering the methodology.

$$n_{ii,T}(0) = n_{0,ii,T}, \quad \forall i \in \mathcal{R}, \quad n_{ij,T}(0) = n_{0,ij,T}, \quad \forall i, j \in \mathcal{R}_{ij}, \quad (20)$$

$$n_i(0) = n_{0,i}, \quad \forall i \in \mathcal{R}, \quad (21)$$

$$n_{i,DQ}(0) = n_{0,i,DQ}, \quad \forall i \in \mathcal{R}, \quad (22)$$

$$n_{ij,Q}(0) = n_{0,ij,Q}, \quad \forall i, j \in \mathcal{R}_{ij}, \quad (23)$$

The states and control inputs are bounded in this problem: First, the accumulations $n_i(t)$ have a lower bound of zero, and the upper bounds are the maximum accumulations $\bar{n}_i$ [aircraft], respectively, see (14). Similarly, the queues $n_{i,DQ}(t)$ and $n_{ij,Q}(t)$ have lower bounds of zero, and the upper bounds are the maximum capacity $\bar{n}_{i,DQ}$ [aircraft] and $\bar{n}_{ij,Q}$ [aircraft], see (15) and (16), respectively. Note that the accumulations $n_{ii,T}(t)$, $n_{ij,T}(t)$ are not bounded directly. In addition, the control inputs $u_{d,i}(t)$, $u_{b,ij}^I(t)$, $u_{b,ij}^O(t)$ have lower bounds $\underline{u}_{d,i}$, $\underline{u}_{b,ij}^I$, $\underline{u}_{b,ij}^O$ [–] and upper bounds $\bar{u}_{d,i}$, $\bar{u}_{b,ij}^I$, $\bar{u}_{b,ij}^O$ [–], see (17)–(19), respectively. Lastly, $n_{0,ii,T}$, $n_{0,ij,T}$, $n_{0,i}$ and $n_{0,i,DQ}$, $n_{0,ij,Q}$ are the initial accumulation and queue values, respectively, see (20)–(23). To conclude, the objective function (13) is minimized subject to the dynamics (4)–(8), the constraints (14)–(19), with the initial values (20)–(23).

2.5. Traffic control strategies

The optimal control problem (4)–(8) and (13)–(23) is solved by applying a Model Predictive Control (MPC) approach, see e.g. Geroliminis et al. (2013). MPC is a form of rolling horizon control in which the current control inputs are obtained by solving a finite-horizon open-loop optimal control problem at each time step, with a current state feedback from the plant being the initial state of the model. The open-loop optimization problem yields a sequence of optimal control inputs after several iterations of solving nonlinear programming, and the first control action in this sequence is applied to the plant, then the procedure is carried out again. This scheme of feedback control, i.e. the feedback loop of states from the plant to the model as initial states for the optimization, can handle errors between the prediction model and the plant.

The MPC controller obtains the optimal control sequence for the prediction horizon N_p by solving an optimization problem formulated with the prediction accumulation-based model, where Δ_k [s] is the time step of the MPC controller. The MPC controllers calculate the optimal control inputs $u_{d,i}(t)$ ($\forall i \in \mathcal{R}$) and $u_{b,ij}^I(t)$, $u_{b,ij}^O(t)$ ($\forall i, j \in \mathcal{R}_{ij}$) to minimize the objective function $J(\star)$, see (13), according to Algorithm 1.

Algorithm 1 Description of the algorithm MPC($\star$)

Require: t_k - Current time step $t_k \in [t_0, t_f]$, t_0 - initial time, t_f - final time, Δt_C - control step.

Require: N_p - prediction horizon.

Require: Δ_k - the time step of the MPC controller.

```

1: function MPC( $\star$ )
2:   Initialize: Set the current states (20)–(23) in step  $t_k$  from plant model;
3:   Initialize: Simulation and control settings;
4:   if  $t_k \bmod \Delta t_C$  then
5:     for  $t_k = t_k, \dots, t_k + \Delta_k \cdot N_p$  do
6:       Predict the future states over the time horizon;
7:       for  $\kappa = 0, \dots, N_p - 1$  do
8:         Solve the optimization problem (13)–(23):
9:         Minimize the cost function (13) with respect to control inputs  $u_{d,i}^I, u_{b,ij}^I, u_{b,ij}^O$ ;
10:        Subject to the system dynamics (4)–(8) and constraints (14)–(19);
11:        Apply the first control inputs  $u_{d,i}^I, u_{b,ij}^I, u_{b,ij}^O$  to the system;
12:        Measure the updated states;
13:        Shift the time horizon by updating the states and controls;
14:       end for
15:     end for
16:   end if
17:   return  $u_{d,i}^I, u_{b,ij}^I, u_{b,ij}^O$ ;
18: end function

```

The optimal control inputs for the macroscopic level should be transformed to the microscopic level, i.e. to obtain the optimal solution for the aircraft level. The transformation is done by utilizing an aircraft command subsystem with its command algorithms, as depicted in Fig. 2. Currently, in conventional airplanes, the aircraft command is applied in reality with human-based control, as the operator in the control center gives the pilot command to change path or velocity. In future LAAT systems, aircraft command can be applied with an unmanned aircraft system, which works automatically and communicates with the autopilot or with the pilot. The aircraft command systems can control the aircraft in matters of routing, speed limit, and others.

In this study, both DC and BC controllers are designed according to the feedback control strategy, then the aircraft command system determines the departure time and waiting time at the boundary of each aircraft, respectively. Both controllers obtain control inputs which are applied on a control step Δt_C in the framework, as shown in Fig. 2. It is clear that the control step Δt_C is larger or

equal than the time interval of MFD variables estimation Δt_M , since these variables are needed as feedback inputs to the controllers. Two command algorithms are determined: one for the departure controller and one for the boundary controller.

First, the optimal departure control inputs $u_{d,i}(t)$ for the macroscopic level are transformed to the microscopic level as waiting times for aircraft departures $\Delta t_{d,i}$ in each region i . In other words, the departure time of aircraft $t_{d,\mathcal{A}}$ will be delayed by $\Delta t_{d,i}$, i.e. $t_{d,\mathcal{A}} \leftarrow t_{d,\mathcal{A}} + \Delta t_{d,i}$. The transformation of the control input $u_{d,i}(t)$ to the waiting time to aircraft departure $\Delta t_{d,i}$ is proportional to the control step Δt_C . Furthermore, $\Delta t_{d,i}$ is bounded such that the departure time cannot be earlier than the original value, i.e. $\Delta t_{d,i} \geq 0$ (non-negative), and cannot be later than the final time t_f , i.e. $t_f - t$. The command algorithm for DC is presented in Algorithm 2.

Algorithm 2 Description of the command algorithm for DC

Require: t_k - Current time step $t_k \in [t_0, t_f]$, t_0 - initial time, t_f - final time, Δt_C - control step.

Require: $\mathcal{A}$ - Aircraft index.

Require: $\mathbb{M}_{DQ,i}$ - Departure queued aircraft index set (departing from region i).

Require: $\mathbb{M}_{ina,i}$ - Inactive aircraft index set (expected to depart from region i).

Require: $S_{\mathcal{A}}$ - Aircraft $\mathcal{A}$ status.

Require: $t_{d,\mathcal{A}}$ - Departure time of aircraft $\mathcal{A}$.

Definition: $u_{d,i}(t_k)$ - Optimal departure control inputs.

Definition: $\Delta t_{d,i}$ - Waiting time to depart in region i .

```

1: if  $t_k \bmod \Delta t_C$  then
2:    $u_{d,i}(t_k) \leftarrow \text{MPC}(\star)$ ;
3:    $\Delta t_{d,i} \leftarrow \min(\max(0, \Delta t_C \cdot (1 - u_{d,i}(t_k))), t_f - t_k)$ ;
4:   if  $0 < \Delta t_{d,i}$  then
5:     for all  $\mathcal{A} \in \{\mathbb{M}_{DQ,i}; \mathbb{M}_{ina,i}\}$  do
6:       if  $t_k \leq t_{d,\mathcal{A}} < t_k + \Delta t_C$  then
7:          $t_{d,\mathcal{A}} \leftarrow t_{d,\mathcal{A}} + \Delta t_{d,i}$ ;
8:          $S_{\mathcal{A}} \leftarrow \text{DEPARTURE QUEUE}$ ;
9:          $\mathbb{M}_{DQ,i} \leftarrow \mathbb{M}_{DQ,i} \cup \{\mathcal{A}\}$ ;
10:         $\mathbb{M}_{ina,i} \leftarrow \mathbb{M}_{ina,i} \setminus \{\mathcal{A}\}$ ;
11:       end if
12:     end for
13:   end if
14: end if

```

Second, the optimal boundary control inputs $u_{b,ij}^I(t), u_{b,ij}^O(t)$ for the macroscopic level are transformed to the microscopic level as a queuing assignment for traveling aircraft at the regional boundary ij for the current control step Δt_C . In other words, the optimal boundary control inputs $u_{b,ij}^I$ determines the ratio of aircraft allowed to pass the regional boundary ij , where the remaining aircraft, i.e. $1 - u_{b,ij}^I$, are assigned to wait at the boundary outside the region (the buffer zone). Then, the optimal boundary control inputs $u_{b,ij}^O$ determines the ratio of aircraft, inside the boundary queues, allowed to re-enter the network and discharge from the boundary queue, where the remaining queued aircraft, i.e. $1 - u_{b,ij}^O$, are assigned to continue waiting at the boundary outside the region. Note that, in the case when the boundary control is activated, i.e. $u_{b,ij}^I < 1$, the strategy is activated when the command system randomly detects that aircraft $\mathcal{A}$ is going to cross the boundary ij , the aircraft gets a command to stay fixed in a specific position outside the network as a temporary waypoint, and gets assigned as a boundary queue until said otherwise, according to the discharge of the boundary queues and feedback control strategy. The command algorithm for BC is presented in Algorithm 3.

To conclude, in this paper, we developed two command algorithms, *departure controller (DC)* and *boundary controller (BC)*, to deploy the proposed integrated departure management and boundary control strategies based on the novel model presented in Section 2.4. This arises an original controller, namely *departure and boundary controller (DBC)*. The three strategies DC, BC, and DBC are compared in a simulation environment presented in Section 3.

3. Simulation

To examine the presented control strategies, the LAAT-Flow simulation tool, see Safadi et al. (2023a), was expanded to include traffic control strategies. The expanded LAAT-Flow simulation is used as the plant model. The setup and settings are described in more detail in Section 3.1. To implement the control strategies, one needs first to identify the airspace state via MFD estimation, where the MFD identification methodology is similar to the one presented in Safadi et al. (2023a). Different control strategies are designed in this paper according to Section 2.5, the controllers DC, BC, and DBC are compared under different demand levels and the results are presented in Section 3.2.

In this paper, the LAAT-Flow simulation tool was expanded to enable feedback control strategies at microscopic and macroscopic levels. The tool has significantly been improved since its previous version, see Safadi et al. (2023a). The latest extension allows for

Algorithm 3 Description of the command algorithm for BC

Require: t_k - Current time step $t_k \in [t_0, t_f]$, t_0 - initial time, t_f - final time, Δt_C - control step.

Require: $\mathcal{A}$ - Aircraft index.

Require: $\mathbb{M}_{\text{act}}$ - Active aircraft index set.

Require: $\mathbb{M}_{\text{BQ},ij}$ - Boundary queued aircraft index set at the regional boundary ij .

Require: $\mathbb{M}_{\text{act},ij}$ - Active aircraft index set detected to cross the regional boundary ij .

Require: $\mathbf{S}_{\mathcal{A}}$ - Aircraft $\mathcal{A}$ status.

Definition: $u_{b,ij}^I(t_k), u_{b,ij}^O(t_k)$ - Optimal boundary control inputs.

Definition: $n_{b,ij}^I(t_k)$ - Number of aircraft assigned to enter the boundary queue.

Definition: $n_{b,ij}^O(t_k)$ - Number of aircraft assigned to re-enter the network and discharge from the boundary queue.

```

1: if  $t_k \bmod \Delta t_C$  then
2:    $u_{b,ij}^I(t_k), u_{b,ij}^O(t_k) \leftarrow \text{MPC}(\star)$ ;
3:    $n_{b,ij}^I(t_k), n_{b,ij}^O(t_k) \leftarrow \text{DETBOUQUEUES}(\star)^{(a)}$ ;
4: end if
5: if  $u_{b,ij}^I(t_k) < 1$  then
6:   for all  $\mathcal{A} \in \mathbb{M}_{\text{act},ij}$  do
7:     if  $0 < n_{b,ij}^I(t_k)$  then
8:        $\mathbf{S}_{\mathcal{A}} \leftarrow \text{BOUNDARY QUEUE}$ ;
9:        $\mathbb{M}_{\text{act}} \leftarrow \mathbb{M}_{\text{act}} \setminus \{\mathcal{A}\}$ ;
10:       $\mathbb{M}_{\text{act},ij} \leftarrow \mathbb{M}_{\text{act},ij} \setminus \{\mathcal{A}\}$ ;
11:       $\mathbb{M}_{\text{BQ},ij} \leftarrow \mathbb{M}_{\text{BQ},ij} \cup \{\mathcal{A}\}$ ;
12:       $n_{b,ij}^O(t_k) \leftarrow n_{b,ij}^O(t_k) - 1$ ;
13:    end if
14:  end for
15: end if
16: if  $0 < u_{b,ij}^O(t_k)$  then
17:   for all  $\mathcal{A} \in \mathbb{M}_{\text{BQ},ij}$  do
18:     if  $0 < n_{b,ij}^O(t_k)$  then
19:       if  $\text{CLEARTOREENTER}(\star)^{(b)}$  then
20:          $\mathbf{S}_{\mathcal{A}} \leftarrow \text{ACTIVE}$ ;
21:          $\mathbb{M}_{\text{act}} \leftarrow \mathbb{M}_{\text{act}} \cup \{\mathcal{A}\}$ ;
22:          $\mathbb{M}_{\text{BQ},ij} \leftarrow \mathbb{M}_{\text{BQ},ij} \setminus \{\mathcal{A}\}$ ;
23:          $n_{b,ij}^O(t_k) \leftarrow n_{b,ij}^O(t_k) - 1$ ;
24:       end if
25:     end if
26:   end for
27: end if

```

^(a) Function $\text{DETBOUQUEUES}(\star)$ determines the number of aircraft to exit and re-enter the network according to the control model and MPC solution.

^(b) Function $\text{CLEARTOREENTER}(\star)$ verifies that the aircraft can re-enter the network without interrupting neighboring active aircraft with sufficient safety space.

implementing control strategies and modeling of the aircraft's VTOL behavior in a 3D network. The updated tool includes several new features, including simultaneous MFD-based modeling with microscopic modeling, integration with the numerical optimal control toolbox CasADi, see [Andersson et al. \(2018\)](#), advanced aircraft command subsystem that manipulates the departure time and aircraft waypoints in real-time, and new variables and algorithms (as described in Section 2.5). The aircraft object now includes multi-waypoint flight,³ additional aircraft statuses for departure and boundary queues with relative policies, hovering capabilities on region boundaries, and more. The tool has also been enhanced to perform under large-scale networks and high demands. More details about the structure and extension can be found in [Appendix B](#).

3.1. Simulation setup

Different settings and inputs are required to set the computation to perform the traffic analysis, such as aircraft, airspace, and traffic and simulation settings. These inputs are fed to the plant model, the control model, and the controllers. It is possible to conduct

³ Flight trajectories are assigned to aircraft as waypoints that the aircraft needs to achieve for the mission, starting with the origin location and ending in the destination location; in our simulation, each aircraft gets a set of waypoints that need to be achieved, i.e., a multi-waypoint flight.

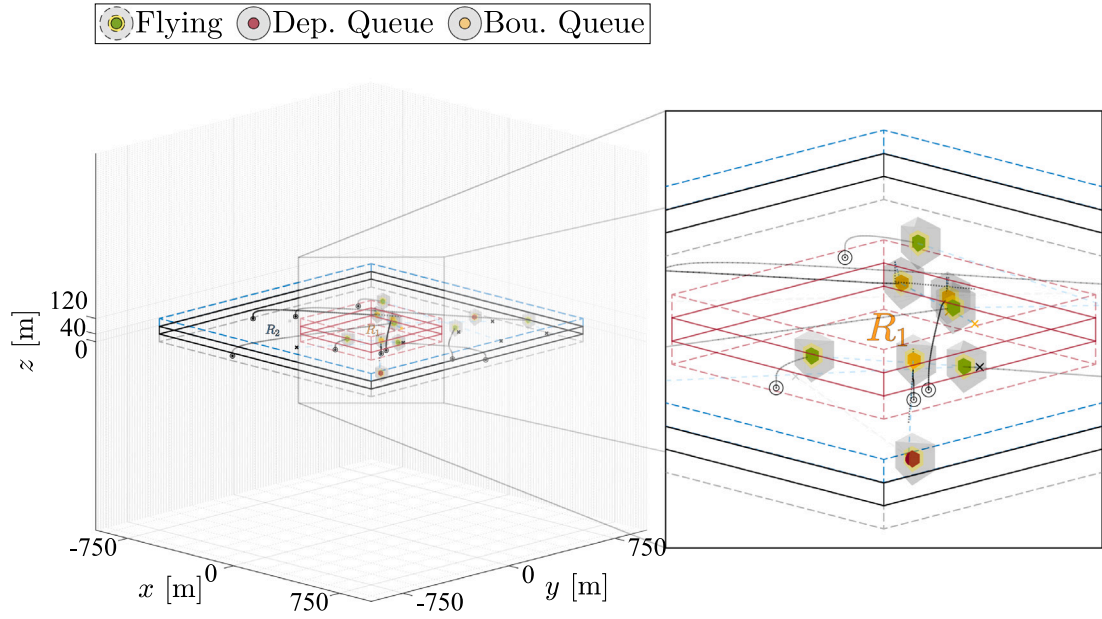


Fig. 3. Airspace setting drawing: airspace network of 1.5 [km] by 1.5 [km] by 120 [m] divided into 3 layers with region 1 (R_1) and region 2 (R_2); including a snapshot from the simulation output of aircraft flying in the airspace (green), departure queues (red), and boundary queues (yellow), with their flight path as a gray curve from origin “o” to destination “x”. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

a wide range of analyzes and studies of the plant model’s outputs, but in this study, our main interest is the aircraft trajectories. The aircraft trajectories are used to construct the MFD variables: flow, density, speed, outflow, production, and accumulation. Additional information extracted from the plant model is used for the aircraft command system, such as the aircraft status (departing, traveling, arriving, etc.) and the aircraft departure time. In the following, we focus on describing the inputs only, as the other parts are well described in Section 2. Note that in Safadi et al. (2023a), different case study examples are simulated to investigate the MFD shapes for LAAT systems, considering different aircraft and airspace settings. Hence, the simulation settings in this study are determined to align with implementing feedback control strategies. In the following, the airspace, aircraft, and traffic and simulation settings are presented, and summarized in Table 1.

3.1.1. Airspace setting

The airspace setting defines the flight’s area and rules, similar to ground traffic networks where the roads and nodes (intersections or roundabouts) are defined. Airspace setting components include the geometry of the area, flight routes, and airspace structure. In this work, we follow the airspace structure presented in Haddad et al. (2021). A 3D network, which allows flights in three dimensional space X–Y–Z, is investigated. Within the network, there is no predetermined structure of flight routes, allowing aircraft to fly in any direction. The network has a size of 1.5 [km] by 1.5 [km] by 120 [m] and is divided into two regions, the city center (region 1) and the periphery (region 2) in each layer. Fig. 3 shows this structure. Each aircraft’s origin and destination are randomly (uniformly) set inside the ground-level network ($z = 0$). The aircraft are designed to model VTOL behavior, meaning they enter and exit the network with vertical takeoff and landing.

3.1.2. Aircraft setting

The aircraft setting determines the properties and parameters for the aircraft dynamics. In the simulation results of this study, the aircraft setting is the same for all aircraft, the maximum speed $v_{m,A}$ is chosen as 20 [m/s], and the maximum acceleration $a_{m,A}$ is chosen as 5 [m/s²]. Moreover, recall that the safety, avoidance, and detection spaces should be determined to actuate the controllers. In this study, the following values are chosen: the safety radius $r_{s,A}$ is 15 [m], the avoidance radius $r_{a,A}$ is 22.5 [m] ($r_{a,A} = 1.5 \cdot r_{s,A}$), and the detection radius $r_{d,A}$ is 45.5 [m].

3.1.3. Traffic and simulation settings

The traffic and simulation settings are defined as follows: (i) the simulation duration t_f is set to 1 [hr] (3600 [s]); (ii) the simulation time step Δt_S is set to 0.5 [s]; (iii) the MFD variables are calculated during a specific time interval $\Delta t_M = 60$ [s]; and (iv) the traffic inflow $q(t)$ [aircraft/s] has a trapezoidal profile that includes warm-up and cool-down periods, with a maximum traffic inflow rate value q_m [aircraft/s].

For the control strategies analysis, additional settings were set to perform and compare the strategies. After evaluating the airspace condition during Δt_M , the control strategies are applied in control step Δt_C and it is chosen as 60 [s].

Table 1

Airspace, aircraft, and traffic and simulation settings.

Airspace setting		Aircraft setting		Traffic and simulation settings	
Parameter	Value and units	Parameter	Value and units	Parameter	Value and units
Δx	1500 [m]	$v_{m,A}$	20 [m/s]	q_m	[0.1, ..., 3.7] [aircraft/s]
Δy	1500 [m]	$a_{m,A}$	5 [m/s ²]	t_f	3600 [s]
Δz	120 [m]	$r_{s,A}$	15 [m]	Δt_S	0.5 [s]
Origin	Uniformly random distributed	$r_{a,A}$	22.5 [m]	Δt_M	60 [s]
Destination	Uniformly random distributed	$r_{d,A}$	45.5 [m]	Δt_C	60 [s]
Paths	Free-flight			Δk	60 [s]
				N_p	5 [-]

In addition, the MPC scheme is built using direct multiple shooting, see e.g. [Bock and Plitt \(1984\)](#), and realized in MATLAB (R2023b) using the numerical optimal control toolbox CasADi ([Andersson et al., 2018](#)), with the solver IPOPT ([Wächter and Biegler, 2005](#)), while the dynamics are discretized with the Euler method with a time step of 60 [s]. The prediction horizon is chosen as $N_p = 5$.

3.2. Implementation of control strategies

Based on the LAAT model and framework, the results for implementing different control strategies, i.e., departure controller (DC), boundary controller (BC), and integrated departure and boundary controller (DBC), are presented and compared with no control (NC) case. Control strategies can be implemented with the knowledge of airspace identification at the network and region levels.

The presented results in this section were obtained for a 3D network case study, wherein a low-altitude aircraft was modeled with VTOL (Vertical Takeoff and Landing) behavior. This means that the aircraft was assigned to take off and land vertically from its origin and destination on the ground. The 3D network case study comprises two separate layers. The first layer is dedicated to the takeoff and landing zone. The second layer is used for flying within the network, and it includes a buffer zone that serves as a space for boundary queues. The purpose of the buffer zone is to prevent any potential collisions or other disruptions that could occur in the interaction between the traveling aircraft and the queueing aircraft.

The developed control strategies are tested and simulated with the same setting with inflow rate $q_m = 2.5$ [aircraft/s]. Note that according to the airspace identification, the estimated maximum outflow (capacity) is $G_m = 2.3$ [aircraft/s], therefore the condition in the system are over-saturated. Additionally, the following information is needed for the implementation: the accumulation lower bounds are zero, and the upper bounds are $\bar{n}_1 = 99$ [aircraft], $\bar{n}_2 = 405$ [aircraft]; the queue lower bounds are zero, and the upper bounds are $\bar{n}_{1,DQ} = 99$ [aircraft], $\bar{n}_{2,DQ} = 405$ [aircraft] and $\bar{n}_{12,Q} = 99$ [aircraft], $\bar{n}_{21,Q} = 405$ [aircraft]; the control input lower bounds are $u_{d,i} = 0$ [-], $u_{b,ij}^1 = 0$ [-], $u_{b,ij}^0 = 0$ [-] and the upper bounds are $\bar{u}_{d,i} = 1$ [-], $\bar{u}_{b,ij}^1 = 1$ [-], $\bar{u}_{b,ij}^0 = 1$ [-]; the critical accumulations are $n_{c,1} = 33$ [aircraft], $n_{c,2} = 135$ [aircraft]; the control reference values are $u_{r,d,i} = 1$ [-], $u_{r,b,ij}^1 = 1$ [-], $u_{r,b,ij}^0 = 1$ [-].

The MPC formulation being presented in this study involves a multi-objective function that is optimized for the control problem at hand. While several techniques exist in the literature for this purpose, see [Deb et al. \(2016\)](#) and [Gunantara \(2018\)](#), the objective function utilized in this research is weighted by three terms, as described in Section 2.4. The calibration process involved an examination of each term separately, which was then followed by normalization using the cost of the joint case. The objective function, see (13), weighting factors are calibrated as follows: for the DC strategy $\omega_{n_i} = 0.0011$ [1/aircraft], $\omega_{u_{d,1}} = 8.7354$ [aircraft], $\omega_{u_{d,2}} = 35.2758$ [aircraft], and $\omega_{u_{b,ij}^1} = \omega_{u_{b,ij}^0} = 0.0000$ [aircraft]; for the BC strategy $\omega_{n_i} = 0.0012$ [1/aircraft], $\omega_{u_{d,1}} = 0.0000$ [aircraft], $\omega_{u_{d,2}} = 0.0000$ [aircraft], and $\omega_{u_{b,ij}^1} = \omega_{u_{b,ij}^0} = 32.0728$ [aircraft]; and for the DBC strategy $\omega_{n_i} = 0.0004$ [1/aircraft], $\omega_{u_{d,1}} = 5.4924$ [aircraft], $\omega_{u_{d,2}} = 22.1800$ [aircraft], and $\omega_{u_{b,ij}^1} = \omega_{u_{b,ij}^0} = 95.4707$ [aircraft]. It is worth noting that the calibration results indicate the boundary control strategy carries greater penalty than the departure control strategy in the DBC case, i.e., $\omega_{u_{b,ij}^1} = \omega_{u_{b,ij}^0} > \omega_{u_{d,i}}$. This is because the boundary queues have more impact on the airspace conditions and consume more energy to hover at the region boundaries, making it essential to prioritize this control strategy to ensure optimal performance. While the calibration of the weights is based on control principles, it could also take into consideration behavioral aspects, such as value of time, waiting vs. moving ratio (as also occurs in public transport literature) and scheduling delays. This can be a future priority for a paper with more emphasis on the acceptability of these services from a user point of view.

In traffic systems, outflow G [aircraft/s], density K [aircraft/m³], and speed V [m/s] are crucial aspects that give us an idea about the network behavior. These macroscopic variables are analyzed through simulation tests with different control strategies and are presented in Fig. 4(a), (b), (c), respectively. The simulation results show that the DBC controller is the most efficient in reducing the airspace density with the same demand, while the NC and BC controllers reach a congestion regime. Additionally, the DBC controller achieves the highest speed profile, while in NC and BC cases, the speed only decreases throughout the simulation.

The simulation results comparing the performances of different traffic control strategies, namely NC, DC, BC, and DBC, for the outflow G [aircraft/s] vs. accumulation n [aircraft] are presented in Fig. 5(a). The simulation results reveal that the integrated DBC outperforms both DC and BC in terms of airspace outflows. The DBC controller ensures that the system remains in a free-flow state with stable behavior even in over-saturated conditions, where DC successfully avoids gridlock but indicates unstable behavior in the system. On the other hand, the NC and BC controllers lead to gridlock conditions due to the high demand level.

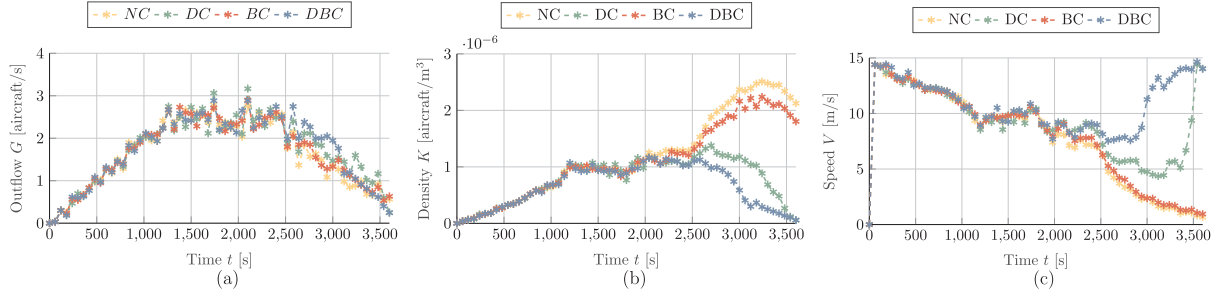


Fig. 4. (a) Outflow G [aircraft/s], (b) Density K [aircraft/m³], and (c) Speed V [m/s] vs. time t [s] for different strategies: No Control (NC), Departure Controller (DC), Boundary Controller (BC), and integrated Departure and Boundary Controller (DBC).

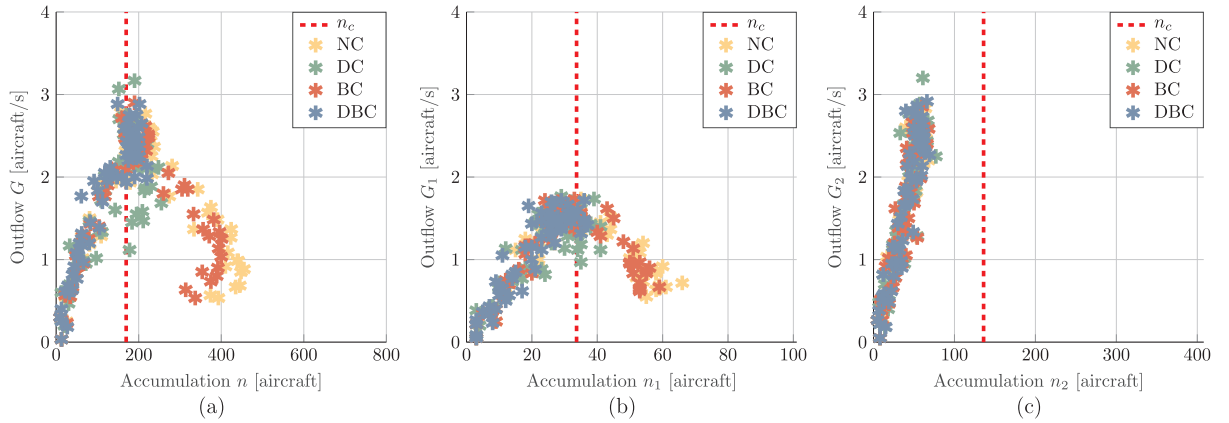


Fig. 5. Outflow G [aircraft/s] vs. accumulation n [aircraft] for: (a) the network level, (b) region 1, and (c) region 2, for different strategies: No Control (NC), Departure Controller (DC), Boundary Controller (BC), and integrated Departure and Boundary Controller (DBC).

The performance of the airspace network is significantly influenced by the partition of the airspace into two regions, i.e., region 1 (city center) and region 2 (periphery). In terms of airspace outflows, i.e., MFD shapes, the region's performance differs for each region. To illustrate, Fig. 5(b) and (c) respectively depict the relationship between the outflow G [aircraft/s] and accumulation n [aircraft] for regions 1 and 2, comparing different control strategies. Interestingly, it is evident that region 2 is consistently in free-flow conditions, whereas region 1 significantly impacts the entire network performance. Similar as presented in Fig. 5(a), the DC and DBC strategies maintain a free-flow condition in region 1, while the NC and BC strategies lead to congestion regimes.

In addition, the states and control inputs in this case study, i.e., Accumulation n_i , Departure Control Inputs $u_{d,i}^I, u_{b,i,j}^O$, Departure Queues $n_{i,DQ}$, Boundary Queues $n_{i,j,Q}$ in regions 1, 2, are presented in Fig. 6. The results demonstrate that the controllers start operating after the warm-up period at $t = 900$ [s] when the accumulation in region 1 approaches the critical accumulation point, i.e., $n_{c,1}$. The MFD-based model predicts the network condition and aims to avoid congestion regime in region 1. The controllers were found to regulate around the critical accumulation points, i.e., maximum outflows, with the DC and DBC controllers performing better than the BC controller. As expected from the calibrated weighting factors, the departure queues were larger than the boundary queues. It is also notable that, despite assigning many aircraft to wait in the boundary, the system in region 1 reaches a larger accumulation than the critical one in the case of the BC controller. Furthermore, the DBC controller outperforms the other controllers as it is activated most of the high-demand period and in the cool-down period (after $t = 2700$ [s]) relaxes the system, unlike the DC controller that still delays aircraft during the cool-down period.

The performance of the MPC controller is affected by the objective function and the related weighting factors, see (13). These weighting factors convert the three terms in the objective function based on the states and control inputs to one criterion $J(\star)$ [aircraft · s], which represents the aircraft travel delays, departure queues delays, and boundary queues delays. It is known in the literature that one can utilize the total time spent (TTS) in the system as a performance index for evaluating and comparing different control strategies. In the presented results, we compare TTS in the system, which is *calculated directly from the plant model results*, by summing the Total Traveled Time (TTT), Total Departure Delay (TDD), and Total Boundary Delay (TBD) from the plant model, therefore, allowing us to evaluate the airspace performance and the controller efficiency.

Considering the aggregated variables for the whole network, simulation results comparing NC, DC, BC, and DBC for TTS are shown in Fig. 7. As shown in Fig. 7, the integrated DBC is more efficient and operational for the LAAT system as it outperforms the other controllers (DC and BC). The DC decreases the TTS by 32% compared with the NC, the BC decreases the TTS by 6.7% compared with the NC, while integrating both controllers decreases TTS by 40% compared with the NC.

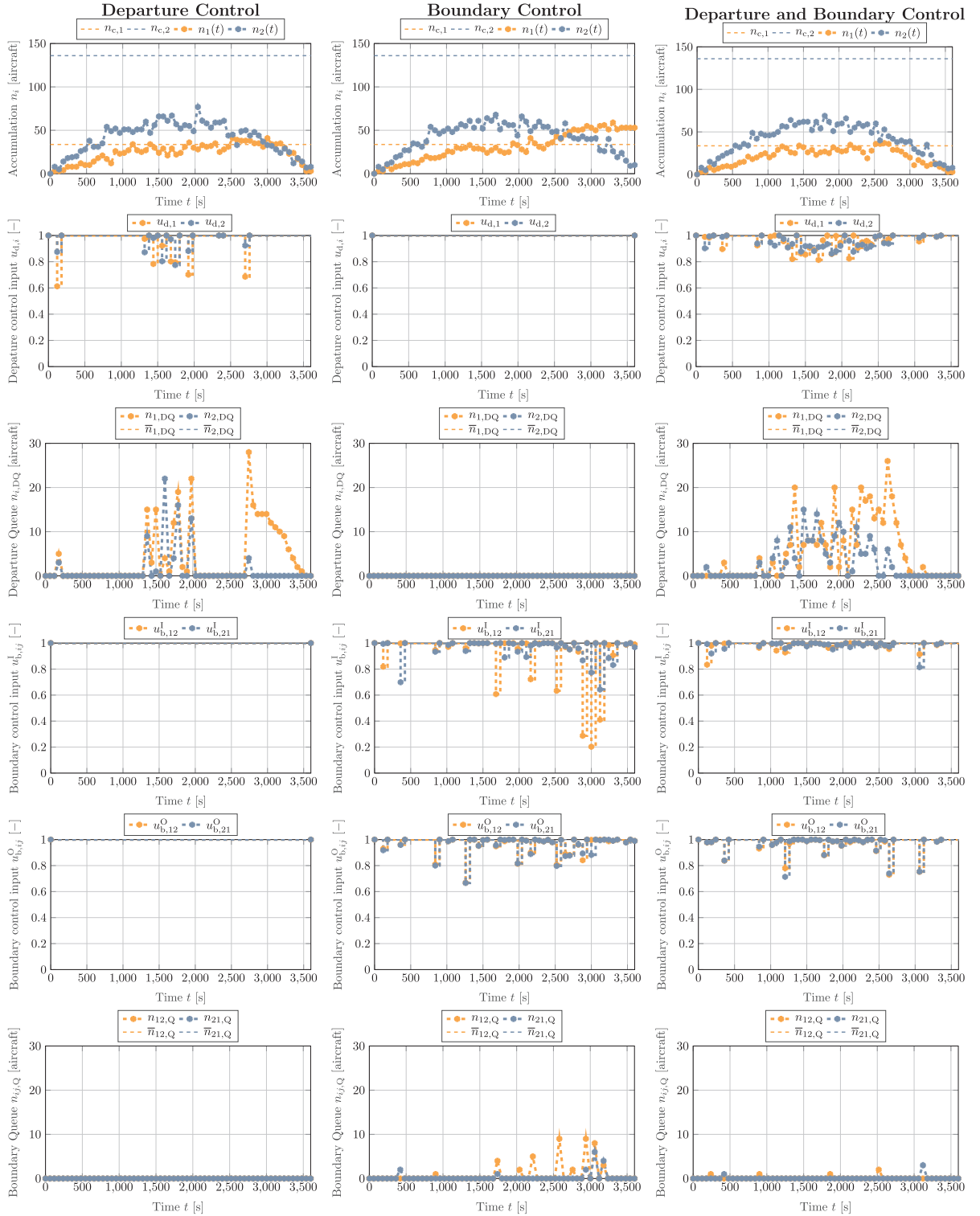


Fig. 6. Accumulation n_i , Departure Control Inputs $u_{d,i}$, Boundary Control Inputs $u'_{b,i,j}$, $u^O_{b,i,j}$, Departure Queues $n_{i,DQ}$, Boundary Queues $n_{i,Q}$ in regions 1, 2 obtained from different strategies: Departure Controller (DC), Boundary Controller (BC), and integrated Departure and Boundary Controller (DBC).

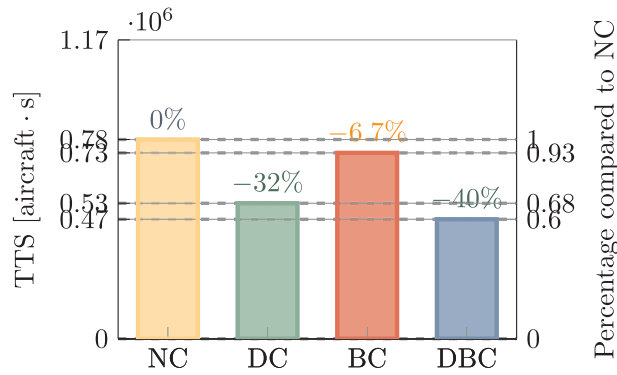


Fig. 7. Total Time Spent (TTS) in the system comparing different strategies: No Control (NC), Departure Controller (DC), Boundary Controller (BC), and integrated Departure and Boundary Controller (DBC).

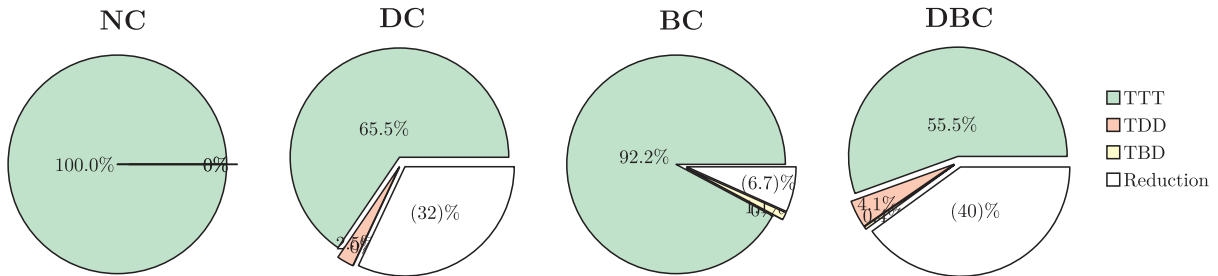


Fig. 8. The percentage of Total Traveled Time (TTT), Total Departure Delay (TDD), Total Boundary Delay (TBD) part of the Total Time Spent (TTS) in the system comparing different strategies: No Control (NC), Departure Controller (DC), Boundary Controller (BC), and integrated Departure and Boundary Controller (DBC). The chart is normalized according to the TTS in the NC case.

The TTS, which is calculated by summing TTT, TDD, and TBD, is further analyzed in Fig. 8. These indicators are normalized by the TTS of the NC case for the different control strategies, i.e., the charts of DC, BC, and DBC cases do not complete to 100% as the TTS is reduced by 32%, 6.7%, 40%, respectively, compared to NC case as also shown in Fig. 7. The study results reveal that the control strategies successfully regulate departure and boundary conditions by imposing relatively small delays on aircraft traveling times. Specifically, when focusing on DC and DBC cases, where the system remains in free-flow conditions, it appears that by increasing the TDD from 2.5% to 4.1% in addition to 0.4% of TBD, the system benefits from an additional 8% reduction in TTS. This fact can also be seen in Fig. 9 where the time series plotting of TTS with the split to TTT, TDD, and TBD is shown for each control step. It is also observed that DBC performs better by activating the departure control strategy earlier compared to DC.

These findings are significant because they suggest that implementing advanced control strategies can make LAAT systems more efficient. By examining the various components of Total Time Spent (TTS), such as Total Traveled Time (TTT), Total Departure Delay (TDD), and Total Boundary Delay (TBD), they can become key performance indicators for evaluating LAAT system performance.

4. Conclusions and future research directions

The main contribution of this paper is developing a new aggregate control-oriented LAAT model which is realized in a novel LAAT framework that (i) models the LAAT traffic flow, and (ii) implements advanced model-based traffic control strategies. The control strategies are actuated through suggested command algorithms to demonstrate the new *integrated departure and boundary control* method.

This paper develops traffic control strategies for LAAT systems by utilizing the MFD concept. An original framework that captures the microscopic and macroscopic levels of LAAT operation was developed. The traffic control strategies are applied at the microscopic and macroscopic levels. The results demonstrate that the total time spent in the system can be decreased by implementing *integrated departure and boundary* control strategy for different traffic conditions, which is aligned with the different UAM Maturity level density scale.

Note that this paper proposes an integrated strategy of departure and boundary control that efficiently balances the demand and supply of LAAT systems. The results suggest that controlling active aircraft has a more significant impact on enhancing the system's efficiency than controlling aircraft waiting to depart. The optimization problem's weight calibration indicates that minimizing boundary queues is crucial compared to departure queues. This is because aircraft at boundary queues are still active while hovering, which consume more energy. Current research, Safadi et al. (2024), is dedicated to further investigating the energy consumption aspect of LAAT systems.

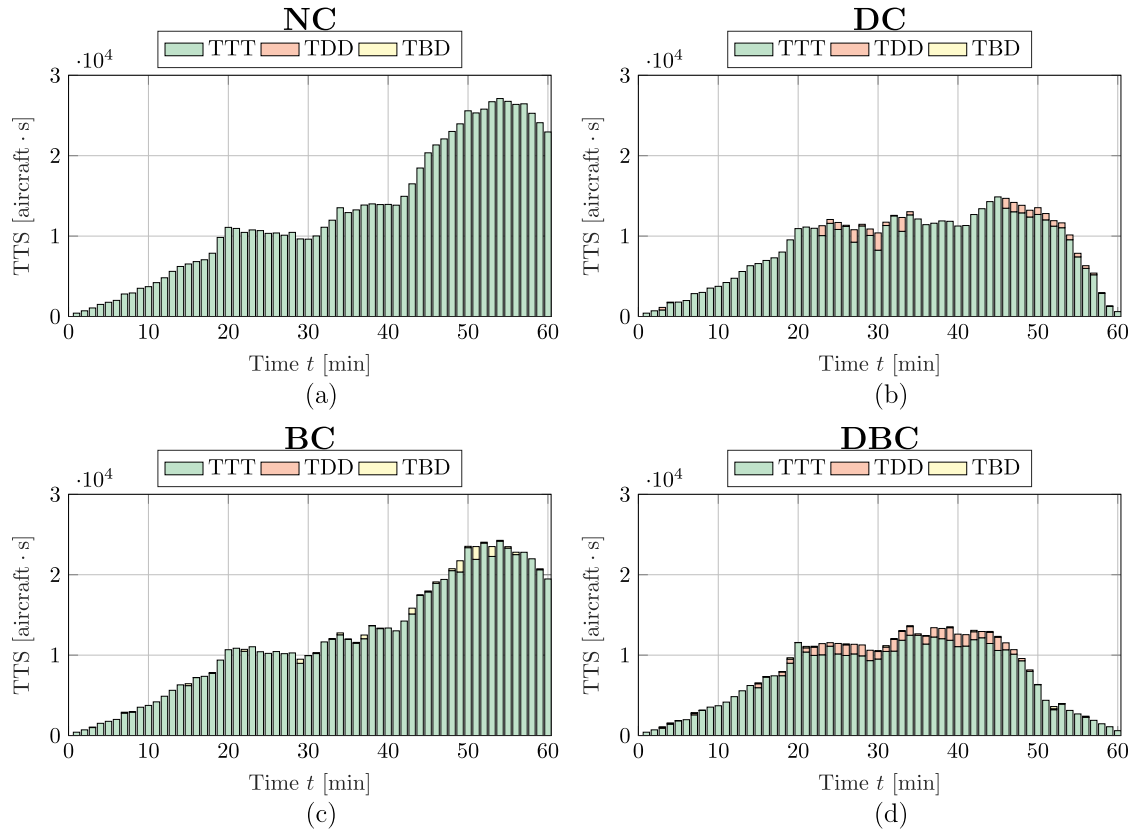


Fig. 9. Time series of Total Time Spent (TTS) for each control step $\Delta t_c = 1$ [min] with the splitting of three terms: Total Traveled Time (TTT), Total Departure Delay (TDD), Total Boundary Delay (TBD), comparing different strategies: (a) No Control (NC), (b) Departure Controller (DC), (c) Boundary Controller (BC), and (d) integrated Departure and Boundary Controller (DBC).

It is important to note that one of the paper limitations is the objective function weighting factor configuration, as it is difficult to calibrate for a larger number of regions. However, the MPC controller in this paper was calibrated to match the simulation settings. Further research is necessary to ensure sufficient configuration or calibration of the weighting factors in the objective function for different scenarios or settings. An additional limitation of controller testing is that the controller's performance is evaluated with a specific demand distribution, and further testing is needed to observe its performance with different demand distributions across regions. Moreover, in this paper, the MPC method was utilized to design the controllers. Considering the combined departure and boundary control problem, future research should focus on testing and comparing the MPC method with other methods from the literature.

It should be stressed that the focus of this paper was on providing a control model for designing departure and boundary control strategies, in such a manner that the genuine behavior of aircraft flow is captured. The control model is an essential component in the development of air traffic control strategies with a hierarchical approach, which determine the optimal decision actuation needed to be deployed for operating efficiently the low-altitude airspace. Note that one method for actuation of the control inputs through the command subsystem at the aircraft (microscopic) level with a specific random queuing setup was introduced to demonstrate the modeling potentials. An additional limitation of the presented case study is the limited airspace structure. Future research should investigate and test other methods and airspace structure setups.

Data availability

Supporting data, code, and the simulation software are available from the authors upon reasonable request via the website [T-SMART \(2024\)](#).

Acknowledgments

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Appendix A. Table of notations

See Table 2.

Appendix B. Traffic simulation structure: Extension of the LAAT-Flow simulation

In the following, we elaborate on the simulation structure that is required to model and control LAAT operation within the extension of the LAAT-Flow simulation, i.e. performing highly detailed analysis for LAAT operation that considers the aircraft dynamics and characteristics, and utilizes control strategies implementation. The LAAT-Flow simulation, Safadi et al. (2023a), has been significantly improved with a new extension. It allows for control strategies and modeling of the aircraft's VTOL behavior in a 3D network. The tool has new features, such as simultaneous MFD-based and microscopic modeling, integration with the numerical optimal control toolbox CasADi, and a more advanced aircraft command subsystem. The aircraft object now includes multi-waypoint flight, additional aircraft statuses, hovering capabilities, and more. The tools can handle large-scale networks and high demands. The simulation is structured based on five elements:

- *Settings* — object contains the settings and parameters;
- *Aircraft* — list of object contains the aircraft information;
- *SimInfo* — object contains matrices for aircraft states, status over time, and additional simulation information;
- *TFC* — object contains the traffic flow modeling variables results and their statistical values.
- *TC* — object contains the traffic control variables results and their statistical values.

A brief description of the framework and its components is provided in Algorithm 4.

Algorithm 4 Description of traffic microsimulation structure

```

1: Settings ← AIRSPACESETTING;
2: Settings ← AIRCRAFTSETTING;
3: Settings ← SIMULATIONSETTING;
4: Aircraft ← INITAIRCRAFTOBJ(★);
5: Settings, TC ← TCINIT(★);
6: for all time  $t$  from initial time  $t_0$  to final time  $t_f$  do
7:   Aircraft, SimInfo, TC ← TCPRE(★);
8:   Aircraft, SimInfo ← AIRCRAFTDEPARTURES(★);
9:   Aircraft, SimInfo ← AIRCRAFTCONTROLLER(★);
10:  Aircraft, SimInfo ← AIRCRAFTMOTION(★);
11:  Aircraft, SimInfo ← AIRCRAFTARRIVALS(★);
12:  Aircraft, SimInfo, TFC ← CALTRAFFICCHARACTERISTICS(★);
13:  Aircraft, SimInfo, TC ← TCPST(★);
14: end for
15: PLOTMOTIONPICTURE(★);
16: EXPORTDATA(★) ⇒ DB;

```

The following functions are set to execute the simulation procedure and model LAAT operation:

- AIRSPACESETTING, AIRCRAFTSETTING, SIMULATIONSETTING — Functions that collect parameters and inputs.
- INITAIRCRAFTOBJ(★) — Function that assigns the aircraft parameters and properties to each *Aircraft* object within a list. Note that additional parameters were added to the original simulation in order to implement the DC, BC, and DBC strategies.
- AIRCRAFTDEPARTURES(★) — Function that assigns the departure time of each aircraft and activates the aircraft to fly if it has permission to depart.
- AIRCRAFTCONTROLLER(★) — Function that determines the aircraft velocity command to arrive at the destination and avoid collision with other aircraft.
- AIRCRAFTMOTION(★) — Function that applies the aircraft motion dynamics according to velocity command and updates the aircraft state.
- AIRCRAFTARRIVALS(★) — Function that determines if the aircraft arrived at its destination and updates the aircraft state.
- CALTRAFFICCHARACTERISTICS(★) — Function that calculates the traffic flow variables and their statistical values.
- PLOTMOTIONPICTURE(★) — Function that plots the virtual environment as a motion picture, with the aircraft information within the airspace and simultaneously the traffic analysis results.
- EXPORTDATA(★) — Function that exports the data and results into a database (DB).

Also, traffic control strategies are integrated within the structure by expanding the existing functions with the following functions:

- TCINIT(★) — Function that initializes the control strategies.

Table 2
Table of notations.

Notation	Units	Description
$\mathcal{A}$	–	Aircraft index
M	–	Aircraft index set
t	s	Current time
$\mathbf{p}_{\mathcal{A}}(t)$	m	Position of aircraft $\mathcal{A}$ at time t
$\mathbf{v}_{\mathcal{A}}(t)$	m/s	Velocity of aircraft $\mathcal{A}$ at time t
$\mathbf{a}_{\mathcal{A}}(t)$	m/s ²	Acceleration of aircraft $\mathcal{A}$ at time t
$v_{m,\mathcal{A}}$	m/s	Maximum velocity of aircraft $\mathcal{A}$
$a_{m,\mathcal{A}}$	m/s ²	Maximum acceleration of aircraft $\mathcal{A}$
$t_{d,\mathcal{A}}$	s	Departure time of aircraft $\mathcal{A}$
$O_{\mathcal{A}}$	m	Origin point of aircraft $\mathcal{A}$
$D_{\mathcal{A}}$	m	Destination point of aircraft $\mathcal{A}$
$\tilde{p}_{D,\mathcal{A}}(t)$	m	Predefined route of aircraft $\mathcal{A}$ at time t
$r_{s,\mathcal{A}}$	m	Safety radius of aircraft $\mathcal{A}$
$S_{\mathcal{A}}$	–	Safety space of aircraft $\mathcal{A}$
$r_{a,\mathcal{A}}$	m	Avoidance radius of aircraft $\mathcal{A}$
$A_{\mathcal{A}}$	–	Avoidance space of aircraft $\mathcal{A}$
$r_{d,\mathcal{A}}$	m	Detection radius of aircraft $\mathcal{A}$
$D_{\mathcal{A}}$	–	Detection space of aircraft $\mathcal{A}$
$\mathbf{v}_{c,\mathcal{A}}(t)$	m/s	Velocity command of aircraft $\mathcal{A}$ at time t
Δt_M	s	Time period of aggregation
$Q(t)$	aircraft/m ² /s	Average absolute aircraft flow at time t
$K(t)$	aircraft/m ³	Generalized aircraft density at time t
$V(t)$	m/s	Generalized speed at time t
$n(t)$	aircraft	Accumulation (number of aircraft) at time t
$G(t)$	aircraft/s	Outflow (trip completion rate) at time t
$P(t)$	aircraft/s	Production at time t
i, j	–	Region index
$\mathcal{R}$	–	Set containing all regions in the network
$\mathcal{R}_{ij}$	–	Set containing all neighboring regions at each region i with destination to another region j
$u_{d,i}(t)$	–	Departure control input for region $i \in \mathcal{R}$ (the ratio of the inflow in each region i)
$u_{b,ij}^{\dagger}(t)$	–	Boundary control input; determines the transfer flow from region i to neighboring region j which applies the queuing behavior at the boundary ij
$u_{b,ij}^O(t)$	–	Boundary control input; determines the released boundary queue flow from region i to neighboring region j
$\dot{n}(t)$	aircraft/s	Time derivative of accumulation in network
$d(t)$	aircraft/s	Inflow rate (demand) in network
$G(n(t))$	aircraft/s	Outflow rate (supply) in network
n_c	aircraft	Critical accumulation in network (the number of aircraft that allows flying freely)
G_m	aircraft/s	Maximum outflow point in network
a_1, a_2, a_3	s, –, aircraft	Estimated parameters from curve fitting
$u_{b,ij}(t)$	–	Control input which determines the ratio of the outflow rate in region i with destination to neighboring region j
$G_i(n_i(t))$	aircraft/s	Outflow rate (supply) in region i
$b_{ij}(t), b_{ii}(t), b_{ji}(t), b_{ji}^Q(t)$	–	Weighting factor
$n_{ij}(t)$	aircraft	Accumulation of aircraft at time t in region i with destination to neighboring region j
$\dot{n}_{ij}(t)$	aircraft/s	Time derivative of accumulation of aircraft at time t in region i with destination to neighboring region j
$d_{ij}(t)$	aircraft/s	Inflow rate (demand) at time t in region i with destination to region j
$n_{ij,T}(t)$	aircraft	Accumulation of <i>traveling</i> aircraft at time t in region i with destination to neighboring region j
$\dot{n}_{ij,T}(t)$	aircraft/s	Time derivative of accumulation of <i>traveling</i> aircraft at time t in region i with destination to neighboring region j
$n_{ii,T}(t)$	aircraft	Accumulation of <i>traveling</i> aircraft at time t in region i with destination i
$\dot{n}_{ii,T}(t)$	aircraft/s	Time derivative of accumulation of <i>traveling</i> aircraft at time t in region i with destination i
S_j^Q	aircraft/s	Saturation flow at the boundary in region j
$n_i(t)$	aircraft	Total accumulation of traveling aircraft in each region i
$\dot{n}_i(t)$	aircraft	Time derivative of total accumulation of traveling aircraft in each region i
$n_{i,DQ}(t)$	aircraft	Departure queue at time t in region i
$\dot{n}_{i,DQ}(t)$	aircraft	Time derivative of departure queue at time t in region i
$n_{i,j,Q}(t)$	aircraft	Boundary queue at time t in region i with destination to neighboring region j
$\dot{n}_{i,j,Q}(t)$	aircraft	Time derivative of boundary queue at time t in region i with destination to neighboring region j
$J(\star)$	aircraft · s	Objective function
J_1	aircraft · s	Objective function term for traveling aircraft delays
J_2	aircraft · s	Objective function term for departure queue delays
J_3	aircraft · s	Objective function term for boundary queue delays
ω_{n_i}	1/aircraft	Weighting factor for J_1
$\omega_{u_{d,i}}$	aircraft	Weighting factor for J_2
$\omega_{u_{b,ij}^{\dagger}}, \omega_{u_{b,ij}^O}$	aircraft	Weighting factor for J_3

(continued on next page)

Table 2 (continued).

Notation	Units	Description
$\bar{n}_i$	aircraft	Maximum accumulations in region i (upper bounds)
$\bar{n}_{i,DQ}$	aircraft	Maximum departure queues capacity in region i (upper bounds)
$\bar{n}_{ij,Q}$	aircraft	Maximum boundary queues capacity in region i with destination to neighboring region j (upper bounds)
$u_{-d,i}^l$	–	Departure control input lower bounds in region i
$u_{b,ij}^l$	–	Boundary control input lower bounds at the boundary ij
$u_{-d,ij}^o$	–	Boundary control input lower bounds at the boundary ij
$\bar{u}_{d,i}^l$	–	Departure control input upper bounds in region i
$\bar{u}_{b,ij}^l$	–	Boundary control input upper bounds at the boundary ij
$\bar{u}_{-d,ij}^o$	–	Boundary control input upper bounds at the boundary ij
$n_{0,i,T}$	aircraft	Initial traveling accumulation in region i with destination i
$n_{0,ij,T}$	aircraft	Initial accumulation in region i with destination to neighboring region j
$n_{0,i}$	aircraft	Initial accumulation in region i
$n_{0,i,DQ}$	aircraft	Initial departure queues in region i
$n_{0,ij,Q}$	aircraft	Initial boundary queues at the boundary ij
N_p	–	Prediction horizon of the MPC controller
Δ_k	s	Time step of the MPC controller
t_k	s	Current time step
t_0	s	Initial time
t_f	s	Final time
Δt_C	s	Control step
$\Delta t_{d,i}$	s	Waiting times for aircraft departures in each region i
$M_{DQ,i}^l$	–	Departure queued aircraft index set (departing from region i)
$M_{ina,i}^l$	–	Inactive aircraft index set (expected to depart from region i).
S_A	–	Aircraft A status
$u_{d,i}^l(t_k)$	–	Optimal departure control inputs
M_{act}^l	–	Active aircraft index set
$M_{BQ,ij}^l$	–	Boundary queued aircraft index set at the regional boundary ij
$M_{act,ij}^l$	–	Active aircraft index set detected to cross the regional boundary ij
$u_{b,ij}^l(t_k), u_{b,ij}^o(t_k)$	–	Optimal boundary control inputs
$n_{b,ij}^l(t_k)$	aircraft	Number of aircraft assigned to enter the boundary queue
$n_{b,ij}^o(t_k)$	aircraft	Number of aircraft assigned to re-enter the network and discharge from the boundary queue
Δt_S	s	Simulation time step
$q(t)$	aircraft/s	Traffic inflow
q_m	aircraft/s	Maximum traffic inflow rate value

- TCPRE(★) — Function that executes command strategies at pre-computation of simulation step, e.g. the command algorithm DC (Algorithm 2).
- TCPPost(★) — Function that executes command strategies at post-computation of simulation step, e.g. the command algorithm BC (Algorithm 3).

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